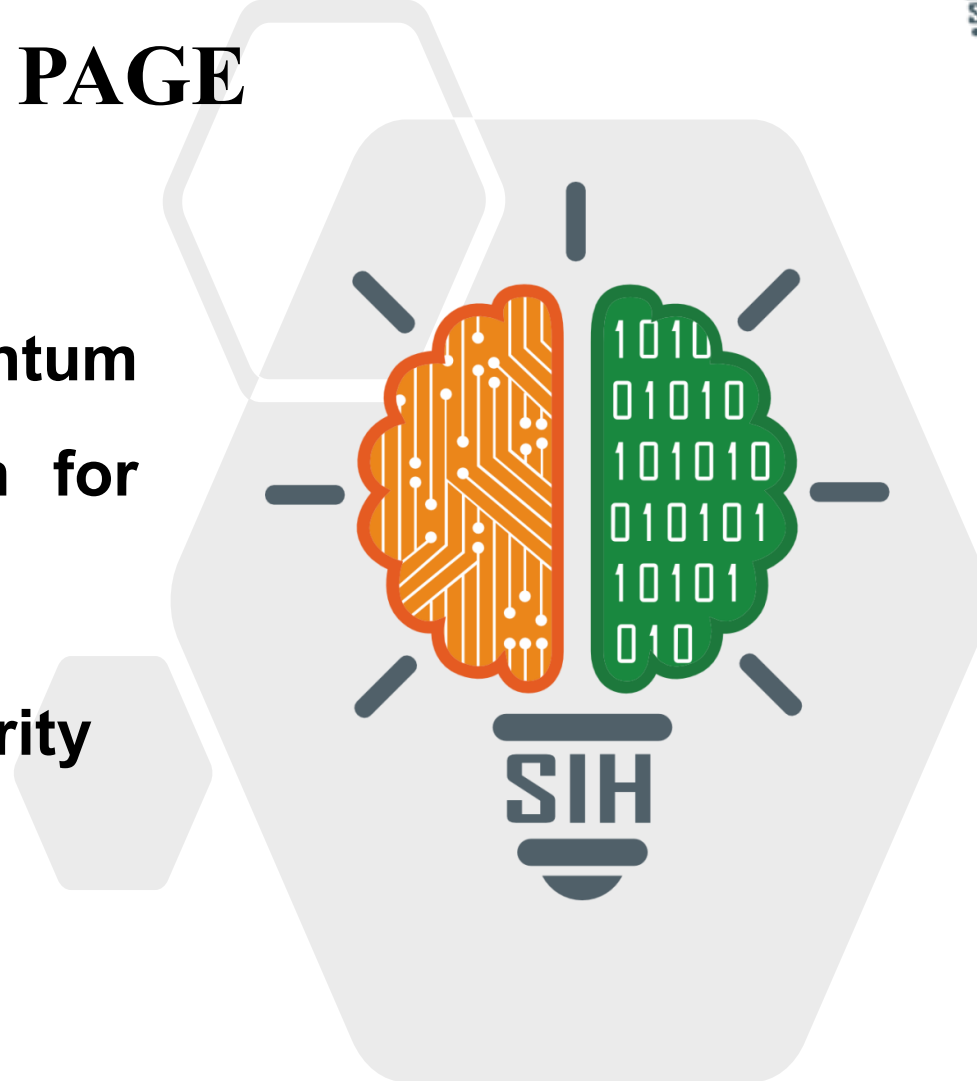


SMART INDIA HACKATHON 2025



TITLE PAGE

- Problem Statement ID – SIH26141
- Problem Statement Title – Quantum Inspired Cyber Threat Detection for DSS
- Theme – Blockchain & Cybersecurity
- PS Category - Software
- Team ID – SIH26-A0H-T004
- Team Name - ARISTELLA



Problem:

- **Quantum Digital Signatures (QDS)** provide a strong foundation for secure authentication, but practical deployments still need to detect **forgery, impersonation, replay and quantum-channel manipulation**.
- Conventional signature verification determines whether a signature satisfies protocol conditions, but does not by itself provide a dedicated **statistical threat-monitoring layer** for anomalous behaviour.

Our Idea:

- Our framework adds a **non-ML statistical security layer** to teleportation-based QDS. It analyses **Pauli-basis measurement outcomes**, compares observed behaviour against legitimate baselines, and uses **statistical thresholds and protocol evidence** to identify suspicious activity and attribute likely attacks.
- The underlying QDS security model remains unchanged; our framework provides the monitoring and detection layer around it.

Proposed Solution:

- **Teleportation-based QDS:** Uses Bell-state entanglement, quantum teleportation and Pauli feed-forward correction as the protocol foundation.
- **Attack Simulation:** Injects **forgery, impersonation, replay, unauthorized verification and channel manipulation** scenarios into controlled QDS sessions.
- **Quantum Measurement:** Performs **X/Y/Z projective measurements** to capture basis-wise behaviour and measurement mismatch.
- **Statistical Threat Engine:** Uses **measurement statistics, confidence bounds and calibrated thresholds** to distinguish normal, suspicious and malicious behaviour.
- **Security Analytics:** Estimates **forgery probability, false acceptance probability and honest abort probability** for measurable security evaluation.



Innovation/Uniqueness:

• QDS-Specific Threat Detection

Designed around teleportation-based Quantum Digital Signature workflows rather than generic network anomaly detection.

• Non-ML & Explainable

Threat decisions are based on **measurable quantum statistics and explicit thresholds**, not opaque model predictions.

• Five Attack Classes

Forgery, impersonation, replay, unauthorized verification and quantum-channel manipulation.

• Evidence-Based Attribution

Combines **measurement behaviour + protocol evidence** instead of treating every failed verification as an attack.

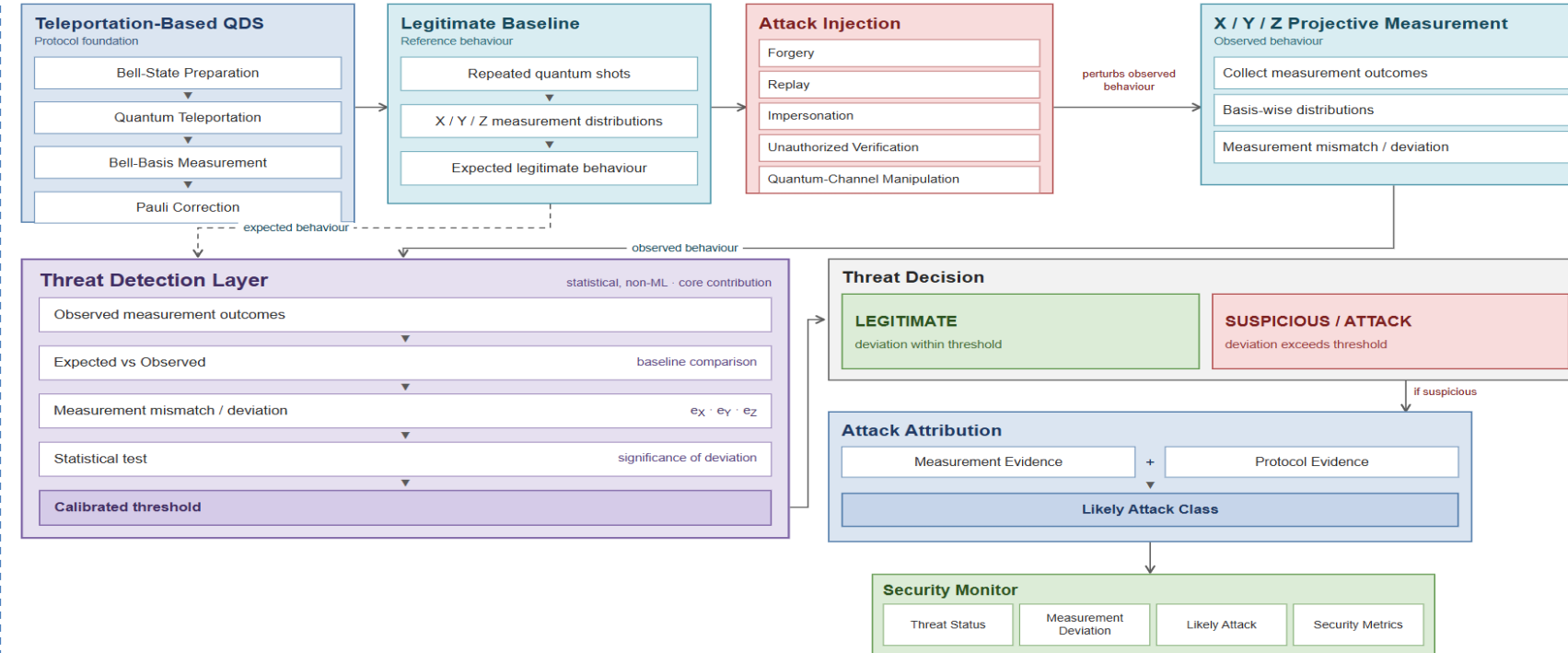
• Security-Aware Metrics

Evaluates **forgery probability, false acceptance and honest abort behaviour**.

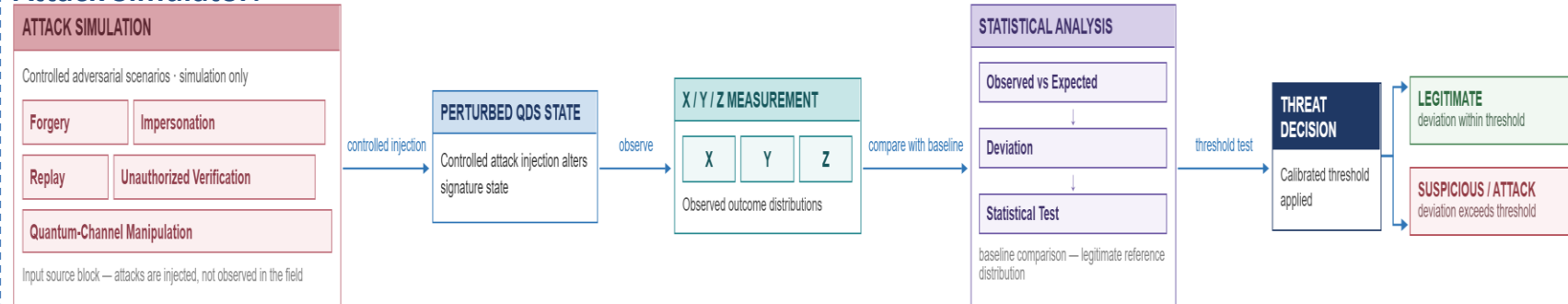
Technology Stack:

- **Qiskit / Qiskit Aer:** Simulates teleportation-based QDS circuits, Bell-state preparation, quantum measurements, and controlled noise/attack scenarios.
- **Python:** Core programming language for attack simulation, measurement processing, statistical analysis, and threat-detection logic.
- **NumPy / SciPy:** Processes X/Y/Z measurement outcomes, calculates mismatch statistics, confidence measures, and statistically calibrated detection thresholds.
- **FastAPI:** Provides backend APIs connecting the quantum simulation and threat-detection engine with the monitoring interface.
- **React / Plotly:** Builds the security-monitoring dashboard and visualizes measurement distributions, deviations, threat status, and attack evidence.
- **SQLite / PostgreSQL:** Stores experiment configurations, measurement results, detection decisions, and security metrics.
- **CPU-Based Simulation:** Enables the complete MVP to run without dedicated quantum hardware; real QPU validation can be added later.

System Architecture:



Attack Simulator:



01 Technical Feasibility

Qiskit / Qiskit Aer — teleportation-based QDS simulation

The complete teleportation-based QDS workflow can initially be simulated in Qiskit/Qiskit Aer, without requiring immediate access to quantum hardware.

Bell-State Preparation

Pauli Operations

X / Y / Z Measurements

02 Computational Feasibility

CPU-compatible MVP — statistical analysis without a dedicated QPU

Measurement processing and the non-ML statistical threat-detection pipeline run on conventional CPU-based computing.

Measurement Mismatch

Statistical Tests

Calibrated Thresholds

03 Deployment Viability

Modular architecture — independently deployable layers

Protocol, attack simulation and detection layers are separable, so the simulator can later be replaced or extended with real quantum hardware or communication testbeds.

Simulation

QPU Validation

Testbed Validation

Challenges:

01 Quantum Simulation Scalability

Challenge — simulation cost

Large-scale quantum circuit simulation becomes computationally expensive as qubit count and measurement shots increase.

Qubit-count scaling

High measurement-shot cost

CPU / memory limitations

02 Realistic Attack Modelling

Challenge — attack fidelity

Simulated attacks must reproduce meaningful changes in QDS measurement behaviour without assuming unrealistic channel conditions.

Forgery & impersonation

Replay & unauthorized verification

Quantum-channel manipulation

03 Baseline & Threshold Sensitivity

Challenge — detection reliability

Legitimate measurement distributions vary with circuit parameters, noise and shot count, making fixed thresholds unreliable.

Legitimate baseline variation

False-positive / false-negative risk

Threshold calibration

MITIGATION

Controlled Simulation

Baseline Calibration

Statistical Threshold

Solutions:

01 Controlled Simulation

Addresses — quantum simulation scalability

Start with compact teleportation-based QDS circuits and controlled shot counts to keep experiments computationally practical.

Qiskit / Qiskit Aer

Controlled circuit size

Configurable shots

CPU-based simulation

03 Calibrated Statistical Detection

Addresses — baseline & threshold sensitivity

Derive thresholds from legitimate measurement variation instead of using arbitrary fixed values.

Legitimate Runs

Baseline Distribution

Statistical Deviation

Calibrated Threshold

Threat Decision

DECISION

Legitimate

Suspicious

02 Attack-by-Attack Evaluation

Addresses — realistic attack modelling

Evaluate each adversarial scenario independently and measure its effect on observable QDS behaviour.

Legitimate Baseline

Attack Injection

Observed Deviation

X / Y / Z Measurement

Attack injection: Forgery · Impersonation · Replay · Unauthorized Verification · Channel Manipulation

CORE SOLUTION

Statistical Threat Detection

Non-ML detection using X / Y / Z measurement behaviour, mismatch statistics and calibrated thresholds.

MEASURE

COMPARE

TEST

DECIDE

ATTRIBUTE

Scalability Roadmap:

SIMULATION

Qiskit / Qiskit Aer
CPU-based MVP

BROADER EVALUATION

More circuits
More attack scenarios
Larger measurement sets

QPU VALIDATION

Real quantum backend validation

QUANTUM TESTBED

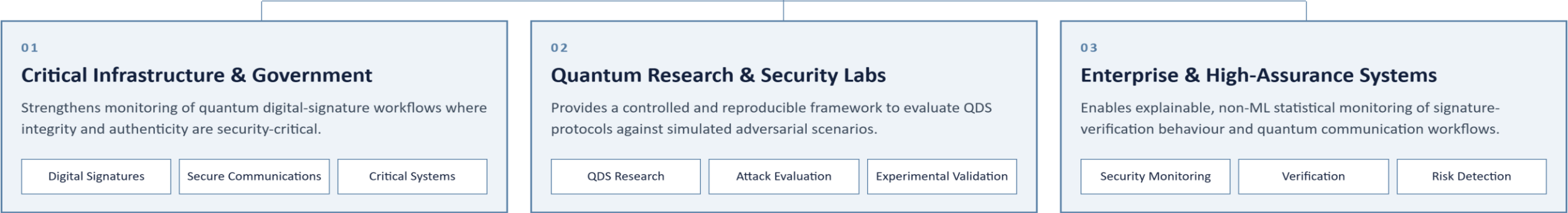
Quantum communication / QDS
testbed integration

MODULAR DETECTION LAYER

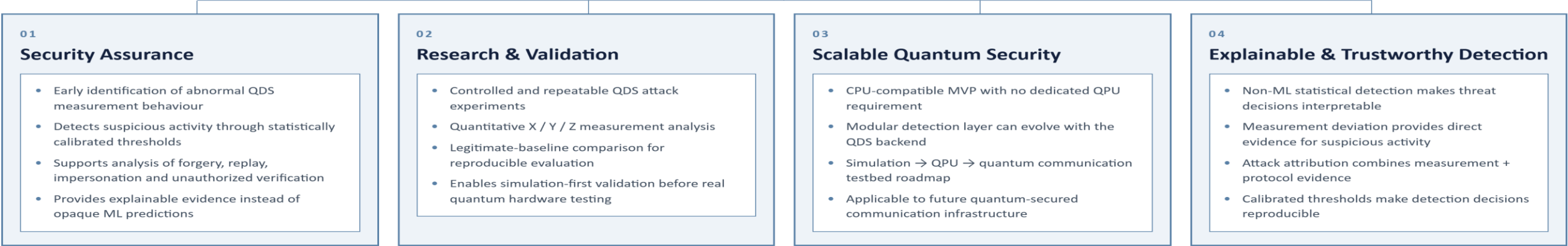
Statistical analysis remains reusable as the underlying QDS execution environment scales.

TARGET USERS & IMPACT

Who benefits → why they benefit → where it applies



BENEFIT ANALYSIS



From quantum measurement behaviour to explainable, statistically grounded threat detection.

Base Paper:

Lu et al., 2022 - <https://www.mdpi.com/1099-4300/24/1/111>

A Verifiable Arbitrated Quantum Signature Scheme Based on Controlled Quantum Teleportation
Entropy, 24(1), 111.

Role: Foundation for the teleportation-based QDS layer.

Related Research:

01 — Gottesman & Chuang, 2001 - <https://arxiv.org/html/quant-ph/0105032v2>

Quantum Digital Signatures

→ QDS foundation

02 — Lee et al., 2022 - <https://ieeexplore.ieee.org/document/9750206>

Eavesdropping Detection in BB84 Quantum Key Distribution Protocols

→ Statistical quantum attack detection

03 — Fiorini et al., 2026 - <https://www.sciencedirect.com/science/article/pii/S1389128626000708>

Threshold-Based Eavesdropper Detection for Partial Intercept-Resend Attack in Noisy BB84 QKD

→ Threshold/statistical detection

04 — Zhao et al., 2023 - <https://www.sciencedirect.com/science/article/pii/S2211379723008112>

Teleportation-Based Continuous-Variable Quantum Digital Signature

→ Teleportation + QDS

05 — Xu & Wang, 2022 - <https://link.springer.com/article/10.1007/s11128-022-03562-1>

A QDS Scheme Based on Superdense Teleportation

→ Alternative teleportation-based QDS

Research Gap:

Existing research provides **teleportation-based QDS protocols** and **statistical attack detection**, mainly in the context of QKD.